

Assignment: 2

Title: Identify commonalities and differences between Publication, Book and Magazine classes. Title, Price, Copies are common instance variables and saleCopy is common method. The differences are, Book class has author and orderCopies(). Magazine Class has orderQty, Currentissue, receiveissue(). Write a program to find how many copies of the given books are ordered and display total sale of publication.

Objective: To learn the concept of polymorphism

Theory:

This section explains the Object Oriented Concept, Polymorphism. Polymorphism is the ability of an entity to behave in different forms. Take a real-world example; the Army Ants. There are different size of ants in the same ant colony with different responsibilities; workers are in different sizes and the queen is the largest one. This is a polymorphic behavior where the entities have a unique feature while they share all other common attributes and behaviors.

Polymorphism is considered as one of the important features of Object Oriented Programming. The word “poly” means many and “morphs” means forms, So it means many forms . Polymorphism allows us to perform a single action in different ways. In other words, polymorphism allows you to define one interface and have multiple implementations..

There are two types of polymorphism in java:

- 1) Static Polymorphism also known as compile time polymorphism
- 2) Dynamic Polymorphism also known as runtime polymorphism

Compile time Polymorphism or Static polymorphism:

Polymorphism that is resolved during compiler time is known as static polymorphism. Method overloading is an example of compile time polymorphism.

Method Overloading: This allows us to have more than one method having the same name, if the parameters of methods are different in number, sequence and data types of parameters .

Example : Method overloading is one of the way java supports static polymorphism. Here we have two definitions of the same method add() which add method would be called is determined by the parameter list at the compile time. That is the reason this is also known as compile time polymorphism.

//Program to explain Method Overloading

```
class SimpleCalculator
{
    int add(int a,int b)
    {
        return a+b ;
    }
}
```

```

        int add(int a,int b,int c)
        {
            return a+b+c;
        }
    }

    public class Demo
    {
        public static void main(String args[ ] )
        {
            SimpleCalculator obj1= new SimpleCalculator();
            System.out.println (obj1.add (10,20));
            System.out.println (obj1.add (10,20,30));
        }
    }

```

Runtime Polymorphism or Dynamic polymorphism

It is also known as Dynamic Method Dispatch. Dynamic polymorphism is a process in which a call to an overridden method is resolved at runtime, that's why it is called as runtime polymorphism. Method overriding is an example of runtime polymorphism.

Method Overriding: Declaring a method in sub class which is already present in parent class is known as Method Overriding. Overriding is done so that a child class can give its own implementation to a method which is already provided by the parent class. In this case the method in parent class is called overridden method and the method in child class is called overriding method.

Method Overriding Example:

We have two classes: A child class Boy and a parent class Human. The Boy class extends Human class. Both the classes have a common method void eat (). Boy class is giving its own implementation to the eat () method or in other words it is overriding the eat () method.

The purpose of Method Overriding is clear here. Child class wants to give its own implementation so that when it calls this method, it prints Boy is eating instead of Human is eating.

Rules of method overriding in Java

Rule #1:

Overriding method name and the overridden method name must be exactly same.

Rule #2:

Overriding method must have the same set of parameters as the overridden method.

Rule #3:

The return type of overriding method name must be same as the super class's method.

Rule #4:

Access modifier of the overriding method must be same or less restrictive than the overridden method's access modifier.

Rule #5:

The overriding method can throw new unchecked exceptions but cannot throw new checked exceptions.

Method Overriding with example (Run time Polymorphism):

```
class Human{
//Overridden method
public void eat()
{
    System.out.println("Human is eating");
}
}
Class Boy extends Human{
//Overriding method
public void eat()
{
    System.out.println("
Boy is eating");
}
public static void main(String args[ ])
{
    Boy obj =new Boy();
//This will call the child class version of eat()
obj.eat();
}
}
```

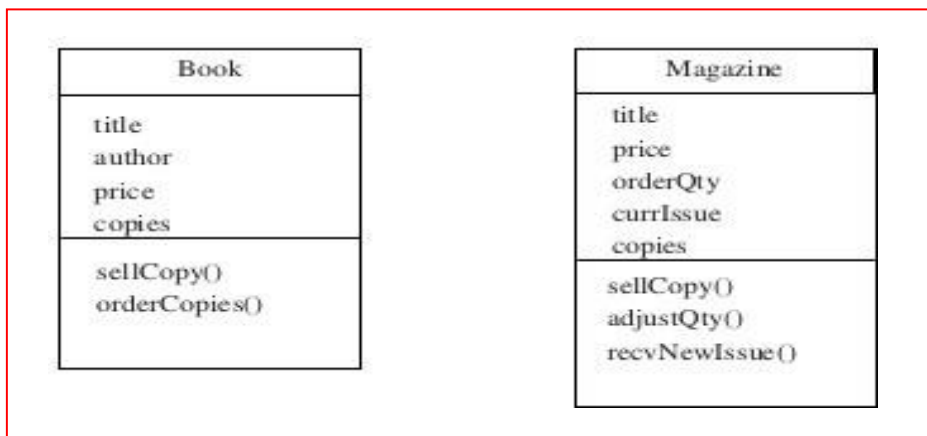
Output:

Boy is eating

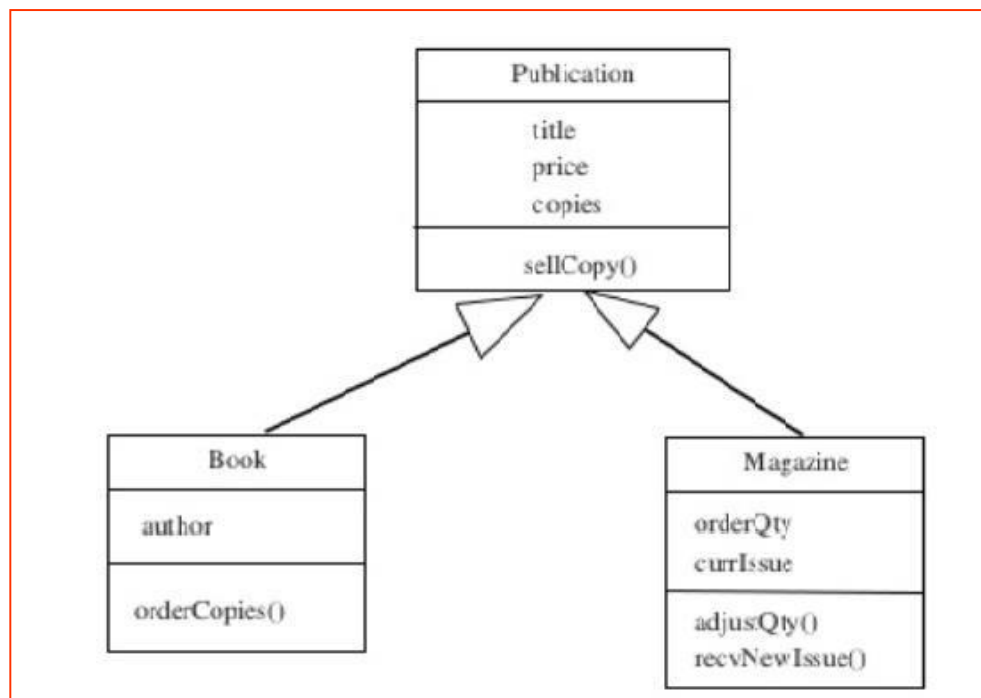
Advantage of method overriding: The main advantage of method overriding is that the class can give its own specific implementation to a inherited method without even modifying the parent class code. This is helpful when a class has several child classes, so if a child class needs to use the parent class method, it can use it and the other classes that want to have different implementation can use overriding feature to make changes without touching the parent class code.

Method Overriding is an example of runtime polymorphism. When a parent class reference points to the child class object then the call to the overridden method is determined at runtime, because during method call which method (parent class or child class) is to be executed is determined by the type of object. This process in which call to the overridden method is .

Consider Book & Magazines both specific type of publication



Attribute title, author & price are obvious parameter. For Book, orderCopies() takes parameter specifying how many copies are added to stock. For Magazine, orderQty is number of copies received of each new issue and currIssue is date/period of current issue. We can separate out these common member of classes into superclass called Publication. The differences will need to be specified as additional member for the 'subclasses' Book and Magazine.



Questions:

1. Define is polymorphism.
2. List out polymorphism types.
3. Differentiate static & runtime polymorphism.
4. Define method overloading.
5. Define method overriding.
6. Differentiate method overloading and method overriding.
7. Explain superclass & subclass.

(write answer for the above questions) & Attached code with output

